

# Multi-task BERT for problem difficulty prediction

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**Abstract**—Existing problem difficulty prediction models are based on professionals' estimation of the difficulty of the problem, or mining relevant feature information from a large number of user records. The recently proposed BERT model is pre-trained on a large unsupervised corpus and has achieved impressive results in various natural language processing tasks. In order to reduce the required feature information and improve the accuracy of problem difficulty prediction, a problem difficulty prediction method based on multi-task BERT (MTBERT) is proposed. Experiments were carried out on the real data sets of LeetCode and ZOJ, and several neural network models were compared to verify the effectiveness of the method.

**Keywords**—Natural language processing; Data mining; BERT; Difficulty prediction

## I. INTRODUCTION

In recent years, with the development of Internet technology, online course education and training platforms such as large-scale development of online courses (MOOC), online Judge systems (OJ). MOOC, like Chinese University MOOC, is maintained by teachers and experts in the corresponding fields. It supports the interaction between students, teachers and teaching assistants. Online judge systems usually have a large number of questions. These online evaluation system platforms provide users with real-time automatic evaluation of submitted solutions. e.g., there are a lot of online tests in Peking University's online judgment system (POJ) and Zhejiang University's online judgment system (ZOJ). Most of these problems are collected from textbooks or the Internet, etc., and the corresponding problem information about the difficulty of these test are often missing. In educational data mining, the difficulty evaluation of problems is an important direction[1]. Understanding text-based learning materials is an important part of an online education system[2]. A large number of online resources make it difficult for users who want to choose the suitable learning material for their current level, e.g., students only want to practice simple programming problems, and it takes a lot of time to find a problem that matches their ability from a large number of problem libraries. Therefore, it is very important to predict the difficulty of the problem in online education. In the literature [2-3], some commonly used neural networks are used to predict the difficulty of the problem, but the model they use does not make good use of specific textual data, and training the model takes too long. To solve these problems, paper proposes a difficulty prediction method based on MTBERT. MTBERT can make full use of related data sets, improve model accuracy, and shorten the time required for training. Experiments show that

MTBERT has significant advantages over other models in terms of language features and problem text information extraction capabilities.

## II. RELATED WORK

Problem difficulty prediction is a long-term research problem in the fields of educational psychology and computer science[4-5]. The current research methods are mainly divided into traditional machine learning methods and deep learning based methods. In document[3], CNN-based architecture is adopted to extract the sentence representations in the problem, and then use the attention strategy to limit the contribution of each sentence to the difficulty of the problem. A test-based training strategy is proposed and the difficulty of the problem is predicted. In document [6] proposed a new comprehensive learning problem representation pre-training method, QuesNet, designed a unified framework to aggregate the problem information and its heterogeneous inputs into a comprehensive vector. On this basis, a two-step pre-training algorithm is proposed to better understand the test questions under unsupervised conditions.

Bidirectional Encoder Representation from Transformers (BERT)[7] model structure is similar to the encoder part of Transformer[8], using a pre-training mechanism. Pre-trained machine learning models have achieved success in various fields, including computer vision and natural language processing. Previous studies show that pre-training on a large corpus can effectively improve the BERT model's ability in classification and other tasks[7]. This training mode can effectively avoid training from scratch, saving a lot of training time.

Multi-task learning, an important method in deep learning[9-10], can effectively share information from the associated supervised data set. Different data sets have different levels of noise, and multi-task learning can provide better generalization ability, improve the model's feature representation ability, and reduce the risk of model overfitting. In document[11], multi-task deep neural network (MT-DNN) based on BERT is proposed. This network uses a large amount of cross-task data to obtain a more general language model.

## III. MULTI-TASK BERT-BASED PROBLEM DIFFICULTY PREDICTION MODEL

In this paper, the difficulty prediction of multi-task BERT is established, which is mainly divided into the following three steps. First, load the pre-training model, initialize the BERT model, and then put the multi-task learning data set into the

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model for training. After the training, save the training results. Finally, put the target training set into the previously trained model for training again. The output of the model is used as the final prediction result.

#### A. BERT

This paper applies the BERT model to the difficulty prediction of the problem. After preprocessing the text information of the title, add the [CLS] token to the head of each text and input it into the BERT model. The BERT model will embed the paragraph information of the text and the position information of the sequence in the text sequence, and then send it to a network similar to Transformer. BERT's self-attention neural network is implemented by stacking Transformer's multi-layer encoder (Encoder), as shown in Fig.1

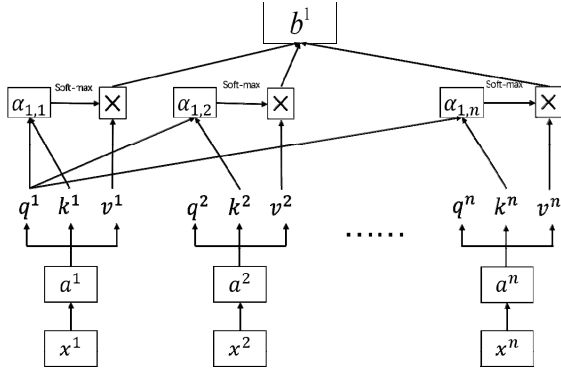


Figure 1. Self-attention neural network.

In the Fig.1:

$$a_{1,i} = \frac{q^1 \cdot k^i}{\sqrt{d}} \quad (1)$$

$$b^1 = \sum_i \text{softmax}(a_{1,i}) v^i \quad (2)$$

The difficulty classification of the problem can be obtained by equations (3)

$$p(\text{Difficulty}) = \text{softmax}(WC) \quad (3)$$

Because the BERT classification task model is used in this paper, the first dimension of the model output corresponding to the [CLS] label in the model input is intercepted as the resulting prediction vector. The vector is passed through a fully connected neural network, and finally the difficulty classification of the problem is obtained, as shown in Fig.2:

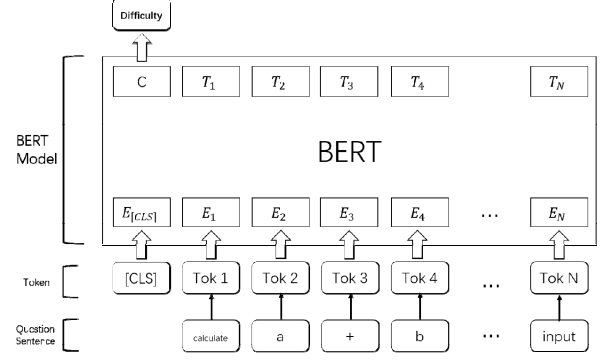


Figure 2. Sequence embedding model and result output.

#### B. Multi-task Learning BERT Model

In the MTBERT model, this paper selects two different sources of data sets for multi-task training, and retains the training model parameters of a single task, so that the learned features can be shared in different data sets as shown in Fig.3:

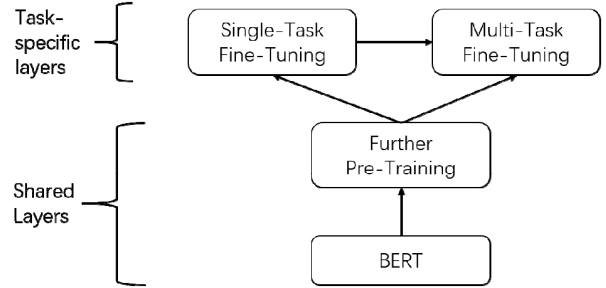


Figure 3. Multi-task learning BERT model.

Because of the online judging system, there may be cheating by the user (the user obtains and submits the answer to the question through online search or other methods). Therefore, the noise of these tasks is relatively large, and the amount of data obtained from the network is limited. It is difficult for the model to extract relevant feature information from the text information of the problem, and multi-task learning can extract the relevant features from other data sets, thereby helping the model to capture useful information.

This article uses two data sets. In order to use the feature information of these two data sets at the same time, so the multi-task fine-tuning of the BERT model, first put the data set used for multi-task fine-tuning into the model for training, here the text uses a smaller learning rate and lower Training epochs to prevent the model from overfitting. After training, save the model data obtained by training. After that, the model after the first training is used to train the target task data set. When training a target task data set, a higher learning rate and training epoch are used to allow the model to fit the data set as much as possible, thereby obtaining a higher prediction accuracy rate.

#### IV. DATA PREPROCESSING

Data preprocessing is a key part of machine learning and deep learning. A good data preprocessing has a great influence on the final result of the model. However, due to the pre-

training form of the BERT neural network language model itself, the model The set does not have very strict requirements. Therefore, when using the BERT mode, only a small amount of modification to the text information of the problem is needed to meet the needs of the model.

#### A. Remove Special Characters

Since the collected data set comes from a web page, the data will carry the label information of the web page, and this information will interfere with the prediction of the result, so these special characters should be deleted during the preprocessing process.

#### B. Convert to Lower Case

The task difficulty prediction task is not sensitive to the case of the text. Before entering the model, the letters of the text information in the data set are converted to lower case format to prevent the problem of letter size format from affecting the prediction results.

#### C. Text Interception

Because the maximum text sequence length that the BERT model can accept is 512. Therefore, when applying the BERT model to the task difficulty prediction task, in order to obtain more information in the problem, the problem text needs to be intercepted into a sequence of length 512. There are usually three interception methods:

Retain the first 510 characters of the text (plus the BERT model's [CLS] and [SEP] token length is 512).

Retain the last 510 characters of the text.

Retain the first 128 characters and the last 382 characters of the text.

According to the data set selected in this article, it is found that there is some background knowledge description in the head of the problem that is irrelevant to the difficulty information of the problem. A large amount of information related to the difficulty of the problem is concentrated in the second half of the text. The combination of the head text plus the tail text of the article.

### V. EXPERIMENT

The main configuration information of the experimental environment in this article is as follows, CPU: Intel(R) Xeon(R) CPU @ 2.30GHz; GPU: Tesla K80; memory: 16G; operating system: Ubuntu 18.04.3 LTS. Use Google's open source deep learning framework TensorFlow to build all neural network models, and interactive training and testing on Jupyter.

#### A. Experimental Data

This article uses two problem difficulty prediction data sets to train and evaluate the model. One of the data sets is only used for multi-task learning and is used for fine-tuning. The first data set comes from the ZOJ English algorithm programming question of Zhejiang University, and contains a total of 3091 data. The second data set comes from LeetCode English algorithmic programming questions and contains a

total of 1657 data. The above two data sets are divided into training set and test set according to the ratio 8:2. The experimental data is shown in Table I.

TABLE I. DATA SET DESCRIPTION

| Attribute | Value                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------|
| Index     | 1                                                                                                                |
| Title     | A + B Problem                                                                                                    |
| Content   | Calculate a + b Input The input will consist of a series of pairs of integers a and b, separated by a space..... |
| Ratio     | 51.01%                                                                                                           |
| Submit    | 5143                                                                                                             |
| Accepted  | 10082                                                                                                            |

Fig.4 is the kernel density distribution map of the difficulty label and the pass rate manually marked in the data set. For the more difficult problems, the pass rate distribution is concentrated in the lower area, and the lower difficulty problems are the pass rate concentrated in the higher area.

Therefore, this paper uses the pass rate of the problem to distinguish the difficulty of the problem. The target with the low pass rate is recorded as the difficult problem, and the target with the high pass rate is recorded as the simple problem.

Set a median to divide the data set, so that the number of difficult problems is the same as the number of simple problems, and when doing random division of the data set, choose a suitable random seed to make the distribution of the training set and the test set balanced.

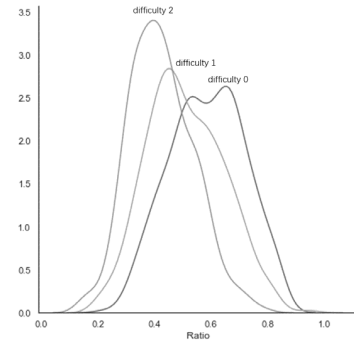


Figure 4. Kernel density of problem passing rate corresponding to different difficulty values.

#### B. Experimental Method

During the experiment, in order to make the model proposed in this paper achieve the best results, the model undergoes multiple hyperparameter selection and optimization, and records the results of each experiment.

The hidden layer size of the BERT model used in this paper is 768 dimensions, 12 attention heads, the training batch size is 4, the maximum sequence length is 512, the learning rate is  $5e-5$ , and the warm-up proportion is 0.1. In this paper, the values of epoch are set to 1, 2, 3, 4, 5, respectively, and the best model on the test set is saved and analyzed.

### C. Results and Analysis

The experimental results are shown in Table II.

TABLE II. RESULTS OF THE DIFFICULTY PREDICTION

| Prediction model | Accuracy      | F1-Score      | Total time(s) |
|------------------|---------------|---------------|---------------|
| BiLSTM           | 0.7049        | 0.6851        | 7329          |
| BERT             | 0.7250        | 0.6909        | 3875          |
| MTBERT           | <b>0.7573</b> | <b>0.7440</b> | <b>3498</b>   |

Table II is the difficulty prediction result of the ZOI data set. The experiment compares the model proposed in this paper with the comparison model on different evaluation indicators. Experimental results show that, on this data set, the method proposed in this paper has a significant improvement in accuracy and F1 value compared with the traditional neural network model and the basic BERT model, which is 4% to 5% higher than BiLSTM, 3% to 4% higher than the basic BERT model on average.

Since there are fewer epoch of convergence, it takes less time to reach the best model. General neural network needs more than 10 epoch to converge, ordinary BERT needs 4 epoch, MTBERT only needs 3 epoch to achieve a more ideal effect.

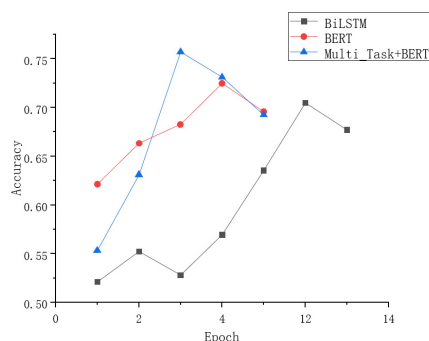


Figure 5. Accuracy of different training epoch.

The model proposed in this paper is obviously superior to the traditional BiLSTM in terms of model convergence speed. This shows that the ordinary neural network model has a weak ability to capture the text information in the problem, so the training round and time are longer. MTBERT has improved these existing problems and achieved a relatively satisfactory result.

### VI. CONCLUSION AND FUTURE WORK

The main work of this paper is to solve the problems of the traditional neural network model in predicting the difficulty of

the problem, such as low accuracy, long training time and complicated preprocessing process. A problem difficulty prediction model based on multi-task BERT is proposed. Comparative experiments show that the proposed method is superior to traditional deep learning methods in accuracy and running time. It shows that the model based on multi-task BERT can learn the feature information of text data to a greater extent, and improve the expression ability of language models. This paper has improved the accuracy of the prediction of the difficulty of the problem, but in the text information processing, there is still a lack of more characteristic feature extraction methods. In future research, will try to compare the difference in the ability to capture the text information of different language models from a theoretical perspective, and choose a more advanced pre-trained language model.

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